

REED PUTNAM PRACTICE · FALL 2026

Polynomials

Week 3 · roots, coefficients, and the degree bound

Thu Oct 8 / Fri Oct 9

The toolbox (1)

Root counting

A **nonzero** polynomial of degree n has at most n roots. So: degrees $\leq n$ agreeing at $n + 1$ points are equal; **infinitely many roots** means $p \equiv 0$. Factor theorem: $p(a) = 0 \iff (x - a) \mid p(x)$.

- **Interpolation**: one polynomial of degree $\leq n$ through $n + 1$ points. Move: an auxiliary polynomial that **vanishes** there, e.g. $(x + 1)p(x) - x = c \prod (x - x_i)$.
- **Integer coefficients**: $(a - b) \mid p(a) - p(b)$. If $p(m) = \pm 1$ often, factor $p \mp 1$: **small** numbers have few divisors.

The toolbox (2)

- > **Vieta:** $x^n + c_{n-1}x^{n-1} + \dots + c_0 = \prod (x - r_i)$ gives $e_k(r) = (-1)^k c_{n-k}$; e.g. $\sum r_i^2 = e_1^2 - 2e_2$. Every symmetric polynomial in the roots is a polynomial in the coefficients.
- > **Degree and growth:** compare degrees and leading coefficients first. $r \mid s \neq 0$ forces $\deg r \leq \deg s$. Formal derivative: $p(x+h) \equiv p(x) + h p'(x) \pmod{h^2}$; repeated root iff $p(a) = p'(a) = 0$.
- > **Locating roots:** real roots by sign changes + IVT; complex roots by the **triangle inequality**; the reciprocal polynomial $x^n p(1/x)$ swaps $|z| < 1$ with $|z| > 1$.

Example 1

Vieta and real roots

If $x^3 + ax^2 + bx + c$ has three real roots, then $a^2 \geq 3b$.

→ work it on the board (3 min), quickly, as a Vieta reminder

Example 2

The integer trick

If $p \in \mathbb{Z}[x]$ takes the value 1 at four distinct integers, then $p(m) \neq -1$ for every integer m .

→ work it on the board (3 min)

Example 3

Infinitely many roots

Find all real polynomials with $p(0) = 0$ and $p(x^2 + 1) = p(x)^2 + 1$.

→ work it on the board (3 min)

How to recognize it

- > “Find all polynomials...” $\Rightarrow$ compare degrees, then **infinitely many roots** of a difference.
- > “Integer coefficients...” $\Rightarrow (a - b) \mid p(a) - p(b)$, or factor $p \mp 1$.
- > “ $p(k) = \dots$ for $k = 0, \dots, n$ ” $\Rightarrow$ an **auxiliary vanishing polynomial**.
- > “Roots real / outside the disk” $\Rightarrow$ sign changes or triangle inequality. Stuck? **Factor** over $\mathbb{C}$.

Pitfalls

- > “ n roots” needs $\mathbb{C}$, multiplicity, $p \neq 0$; bound both degrees before $p = q$; allow $c = 0$.
- > Vieta’s signs alternate: for $x^3 + ax^2 + bx + c$ the sum of roots is $-a$.
- > $2x \mid x^2$ in $\mathbb{Q}[x]$ but not in $\mathbb{Z}[x]$: for integer quotients, check the divisor is monic.

Now you

Problems 1–4.

1–2 warm-ups · 3–4 easy Putnam · 5–6 take-home

35 minutes, groups of 2–3

Solutions posted after Friday.